

AGNIBH DASGUPTA

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Education

2021 - 2026 PhD, Computing & Information Science, **University of Nebraska, Omaha**
2018 - 2020 Master of Science, Computer Science, **Utah State University**
2013 - 2017 Bachelor of Technology, Computer Science, **West Bengal University of Technology**

Publications

- 2025 **Watermarking Language Models through Language Models (IEEE TAI)** [🔗](#)
• Built a prompt-based **LLM watermarking** framework without modifying model weights
• Achieved **88%+ watermark detection accuracy** across 25 **open-source LLMs**
- 2024 **Book Chapter, Imaging Science** [🔗](#)
• Improved the scalability of our attention-based watermarking under heavy compound noise
- 2024 **Leveraging Artificial Intelligence (AI) to Enhance CS Instruction (FIE)** [🔗](#)
• Utilized multi-modality in 3D CNN models for mood detection in K-12 classroom videos
- 2023 **Robust Image Watermarking based on Cross-Attention and ID Learning (CSCI)** [🔗](#)
• Designed an attention-based robust image watermarking scheme using **vision transformers**
• Trained a robust invariant domain using **self-supervised learning**
- 2022 **Perspective transformation layer (CSCI)** [🔗](#)
• Proposed a lightweight layer that learns 2D projections of 3D views for multiple viewpoints

Present and Past Research

- PRESENT **Invariant Representation Learning in LLMs for Model Attribution** (*under review*)
• Analyzed features from **10 open-source LLMs** using **PCA** and custom similarity measures
• Found model-specific representation patterns that enable **reliable model attribution**
- PRESENT **Invariant Feature Learning with Auto-Augmentation for Watermarking** (*under review*)
• Combined **CLIP image-text** features with a **learned augmentation module** evaluated on **ImageNet** and **Visual Genome** datasets
• Exceeded state-of-the-art watermark robustness to camera and environmental distortions
- 2025 **Utilization of SAM2 Model for the Segmentation of Retinal Vascular Leakage on Ultra-wide-field Fluorescein Angiography in Non-infectious Retinal Vasculitis (AAO)**
• Trained a classifier on patches of large optic scans to accurately detect areas of leakage
- 2022 **Sentiment Analysis on Twitter data on Electric Vehicles**
• Analyzed sentiment with **lexical** and **Word2Vec-LSTM** approaches on scraped twitter data
- 2020 **Deep Q Learning applied to stock trading** [🔗](#)
• Built a **reinforcement learning trading system** using auto-scraped S&P index data

Reviewer Service

2022 - PRESENT ACCV, Scientific Reports - Nature, Springer Nature

Teaching Experience

2022 - 2023 **Intern Supervisor**, University of Nebraska at Omaha
2023 - 2025 **Graduate Teaching Assistant**, University of Nebraska at Omaha
Courses: Intro to Python, Capstone
2018 - 2020 **Graduate Teaching Assistant**, Utah State University
Courses: Programming Languages, Intro to C++, Intro to AI

Technical Skills

PROGRAMMING & SYSTEMS:	Python, Bash, Linux, MATLAB, Git, JAX, CUDA
DEEP LEARNING FRAMEWORKS:	PyTorch, TensorFlow, Transformers, LLMs, NLP
DISTRIBUTED & OPTIMIZATION:	DeepSpeed, PyTorch DDP, tf.distribute, Slurm/HPC